


WARNING

POTENTIAL IMPACT INJURIES!

THE UNATTACHED END OF THE GAS LINE MAY EXHIBIT WHIPPING MOTION AND COLLIDE INTO OBJECTS AND PERSONNEL DURING GAS LINE PURGE. FIRMLY HOLD THE UNATTACHED END DURING GAS LINE PURGE.


WARNING

POTENTIAL PROJECTILE HAZARD!

FERROMAGNETIC OBJECTS IN A STRONG MAGNETIC FIELD CAN BECOME DANGEROUS PROJECTILES. NEVER BRING HELIUM DEWARS OR GAS CYLINDERS THAT ARE MADE FROM FERROMAGNETIC MATERIAL INTO THE MAGNET ROOM.

- MAKE SURE ALL EQUIPMENT AND TOOLS USED IN THE MAGNET ROOM ARE NONFERROMAGNETIC.
- HELIUM GAS CYLINDER (BOTTLE) CAPS ARE FERROUS AND MUST BE REMOVED BEFORE BRINGING THE CYLINDER INTO THE MAGNET ROOM OF A RAMPED MAGNET.


WARNING

POTENTIAL COLD BURN HAZARD!

EXPOSURE TO CRYOGENS AND CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE. WEAR PROTECTIVE CLOTHING, NONABSORBENT GLOVES AND GOGGLES WHILE PERFORMING THIS PROCEDURE.


WARNING

POTENTIAL MAGNET QUENCH!

BLOWING WARM HELIUM GAS INTO A MAGNET AT FIELD MAY CAUSE THE MAGNET TO QUENCH. DO NOT ALLOW AN EMPTY DEWAR TO BLOW WARM HELIUM GAS INTO A RAMPED MAGNET.


CAUTION

POTENTIAL MAGNET QUENCH

HELIUM GAS FROM THE HELIUM TRANSFER LINE CAN ENTER THE MAGNET WHILE DEWARS ARE BEING CHANGED IF THE CRYOSTAT STINGER IS LEFT IN THE FILL PORT. THIS CAN CAUSE A RAMPED MAGNET TO QUENCH. ALWAYS REMOVE THE CRYOSTAT STINGER FROM THE FILL PORT AND START THE FILL PROCEDURE OVER AGAIN WHEN CHANGING HELIUM DEWARS WHILE FILLING A RAMPED MAGNET.

Minimum Cryogen Levels (Percent Displayed on Cryogen Monitor)							
Magnet	Ramp	Ramp Down	Shim	Mandatory Fill	Fill (250 Liter Dewars at 90% Efficiency)		
					One Dewar	Two Dewars	Three Dewars
1.5T MR450w (HM Series)	90%	75%	> 75% Minimum	60%	90%	78% - 90%	60% - 78%
1.5 T RD Series	80%	70%	> 75% Minimum	60%	90%	78%-90%	60-78%
3.0T MR750w (UA Series)	98% Top-Off at 80%	98% Top-Off at 80%	> 75% Minimum	60%	90%	78% - 90%	63% - 78%
<i>Do NOT operate (scan) magnet below 40% fill level to avoid possible damage from a magnet quench.</i>							